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Do We Really Need External Tools to Mitigate Hallucinations? SIRA: Shared-Prefix Internal Reconstruction of Attribution

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Abstract

Large vision-language models (LVLMs) often hallucinate when language priors dominate weak or ambiguous visual evidence. Existing contrastive decoding methods mitigate this problem by comparing predictions from the original image with those from externally perturbed visual inputs, but such references can introduce off-manifold artifacts and require costly extra forward passes. We propose SIRA, a training-free internal contrastive decoding framework that constructs a counterfactual reference inside the same LVLM by exploiting the staged information flow of multimodal transformers. Instead of removing visual information from the input, SIRA first lets image and text tokens interact through a shared prefix, forming an aligned multimodal state that preserves prompt interpretation, decoding history, positional structure, and early visual grounding. It then forks a counterfactual branch in later transformer layers, where attention to image-token positions is masked. This branch retains the shared multimodal context but lacks continued access to fine-grained visual evidence, yielding a language-prior-dominated internal reference for token-level contrast. During decoding, SIRA suppresses tokens that remain strong without late visual access and favors predictions whose advantage depends on the full visual pathway. Experiments on POPE, CHAIR, and AMBER with Qwen2.5-VL and LLaVA-v1.5 show that SIRA consistently reduces hallucinations while preserving descriptive coverage and incurring lower overhead than two-pass contrastive decoding. SIRA requires no training, external verifier, or perturbed input, and applies to open-weight LVLMs with white-box inference access.

1 Introduction

Large vision-language models (LVLMs) have made rapid progress on multimodal understanding and generation tasks [19, 30, 24, 39, 3]. Despite this progress, they remain vulnerable to hallucination [21, 27, 32, 17]: the model may describe objects, attributes, or relations that are plausible from linguistic context but unsupported by the image. Such errors often occur when visual evidence is weak or ambiguous, allowing high-probability language continuations to dominate token prediction. Hallucination mitigation therefore requires not only stronger visual representations, but also better control over how visual evidence and language priors are integrated inside the model during decoding.

Recent work addresses this problem through contrastive decoding [20, 6, 18, 37, 16, 23, 7, 31, 4, 35]. These methods compare the model's prediction under the original multimodal input with a reference prediction obtained from a degraded, perturbed, or otherwise counterfactual condition, and then favor

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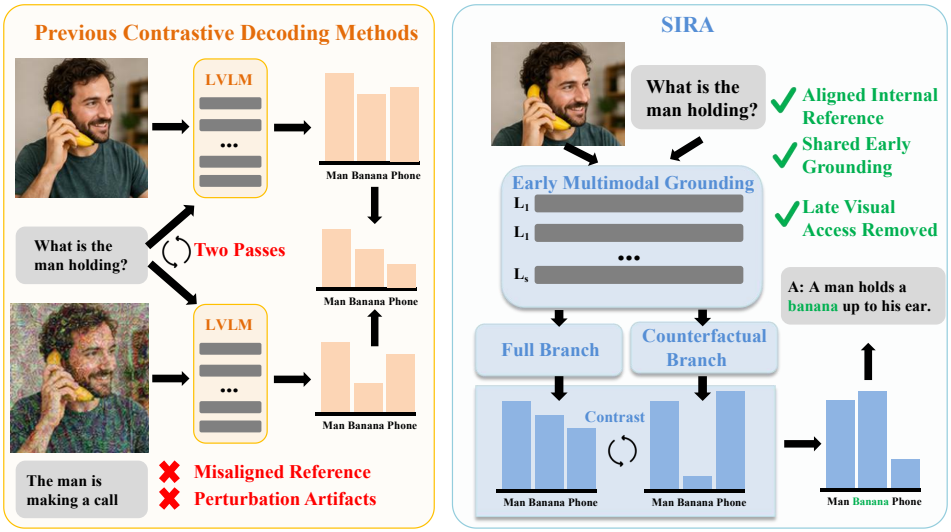


Figure 1: Comparison between representative prior inference-time mitigation methods and SIRA.

tokens whose advantage is attributed to visual evidence rather than generic language plausibility. This token-level view is appealing because hallucinations are ultimately realized through decoding decisions. However, many LVLM-specific contrastive methods construct their reference externally in the input space, for example by blurring, masking, or otherwise weakening the image [18, 37, 7, 4]. This design has two limitations. First, it often requires an additional full forward pass. Second, input-side perturbations may introduce off-manifold artifacts, so the resulting logit discrepancy may reflect perturbation-induced mismatch rather than clean visual attribution [18, 37, 7, 4, 13, 41].

In this paper, we ask a different question: *does contrastive decoding really need an externally constructed counterfactual input?* We argue that the key requirement is an aligned reference that reveals how the same multimodal state would continue when later visual access is restricted. This motivates us to construct the reference by manipulating information flow inside the LVLM. Multimodal transformers integrate image and text information progressively: early layers inject visual evidence into instruction and text-token states, while later layers further use image-token evidence to refine token-level decisions. Therefore, a useful counterfactual branch should not be separated from the input layer, which would destroy early visual-to-text grounding and make the two branches semantically misaligned. It should also not be separated too late, when visual evidence has already been absorbed into hidden states and masking image tokens has little effect.

We propose **S**hared-Prefix **I**nternal **R**econstruction of **A**ttribution (SIRA), a training-free internal contrastive decoding framework for LVLMs. As illustrated in Figure 1, SIRA constructs a synchronized counterfactual branch within the same inference trajectory instead of perturbing the image input. The full and counterfactual branches share the prompt, model parameters, decoding history, positional structure, and early-layer computation. After this shared prefix, the counterfactual branch is prevented from further attending to image-token positions. The shared prefix is therefore not merely an efficiency device: it preserves prompt interpretation and early multimodal grounding, so both branches start from the same image-conditioned semantic state. After branching, the main difference is whether later layers can continue to access fine-grained visual evidence. At each decoding step, SIRA contrasts the logits of the full branch with those of the counterfactual branch. Tokens that remain strong without post-boundary visual access are down-weighted, since they are more likely supported by language priors or generic textual continuations. Tokens whose advantage appears primarily in the full branch are favored as visually grounded. In this way, SIRA performs online internal prior cancellation between two continuations of the same early-grounded multimodal state, without additional training, auxiliary verifiers, external image perturbations, or a second full multimodal forward pass.

Our contributions are summarized as follows:

- We revisit contrastive decoding for LVLM hallucination mitigation from an information-flow perspective, arguing that an effective reference should preserve early multimodal grounding while restricting later image-token access.

• We propose SIRA, a training-free internal contrastive decoding framework that constructs a reference inside the same LVLM via shared-prefix late branching and post-boundary image-token masking.

• We provide comprehensive empirical evidence on POPE, CHAIR, and AMBER across Qwen2.5-VL and LLaVA-v1.5, showing that SIRA consistently reduces hallucination while preserving descriptive coverage and incurring lower overhead than two-pass contrastive decoding.

2 Related Work

2.1 Large Vision-Language Models

Large vision-language models (LVLMs) extend large language models [33, 34, 1] with visual encoders and feature projectors to handle multimodal tasks [19, 30, 24, 39, 3]. While such designs effectively inject visual information into the LLM backbone, how multimodal fusion actually unfolds across transformer layers has only recently been studied. Yin et al. [40] show that shallow layers inject visual information into instruction tokens to form cross-modal semantic representations, while deeper layers intensify intra-visual interactions that aggregate residual features. Kaduri et al. [15] further demonstrate that cross-modal information transfer concentrates in the middle layers, with early and late layers contributing only marginally. Consistent with evidence that factual associations localize to specific transformer layers [26], these findings collectively suggest a *staged fusion pattern*: early-layer visual–textual injection, middle-layer cross-modal consolidation, and a late-layer decision regime where visual evidence has largely been absorbed. Despite this progress, LVLMs still suffer from hallucinations [21, 27, 32, 17] that limit their real-world deployment.

2.2 Hallucination Mitigation in LVLMs

LVLMs produce cross-modal inconsistencies between visual inputs and responses, known as hallucinations [10, 21, 47, 9]. Existing mitigation methods differ mainly in where they introduce the corrective signal. Training-based approaches fine-tune the backbone with hallucination-aware data, preference feedback, or auxiliary objectives [43, 46, 14, 29, 44], but they require additional supervision and model updates. Inference-time methods avoid retraining by manipulating attention, adding external modules, or applying trusted interventions [11, 2, 38, 25, 5, 42, 8, 45]; these methods are easier to deploy than fine-tuning, but they can still add per-step control overhead or rely on components outside the base LVLM.

Contrastive and reference-based decoding methods target the token-selection step more directly. General contrastive decoding and DoLa contrast logits from different generators or layers [20, 6], while LVLM-specific variants build references from perturbed images, instruction changes, self-generated descriptions, frequency-filtered views, masked image heads, self-introspective signals, layer contrasts, object-aligned auxiliary views, or one-layer interventions [18, 37, 16, 23, 7, 13, 31, 4, 35]. These methods share the useful idea that hallucination can be reduced by comparing the original prediction with a reference that exposes language-prior bias. However, the reference is often constructed outside the original inference path or from a condition that is not fully aligned with the original multimodal state, so the logit discrepancy can mix visual attribution with artifacts of perturbation, auxiliary views, or layer mismatch [41]. Motivated by the staged fusion pattern (Section 2.1), we instead construct the reference *inside* the same model: a shared prefix preserves prompt interpretation and early semantic grounding, and post-boundary image-token masking isolates the visual contribution in logit space. Because both branches share weights, prompt, decoding history, and early multimodal interpretation, SIRA avoids introducing a separate off-manifold input condition and yields a more direct token-level attribution proxy than externally constructed references.

3 Method

SIRA augments a single LVLM with a counterfactual branch that shares early-layer computation with the standard multimodal branch and diverges only after a chosen split boundary. The shared prefix preserves semantic alignment: both branches inherit the same prompt interpretation, decoding history, positional structure, and early multimodal grounding. After the boundary, SIRA masks image-token access in the counterfactual branch, so the two branches differ mainly in whether visual evidence can continue to shape token prediction. At every decoding step, SIRA obtains this contrast inside the

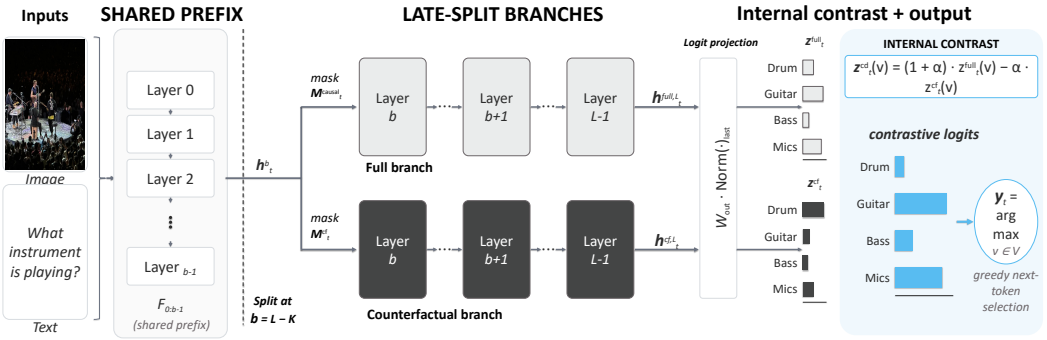


Figure 2: SIRA overview. B in the figure denotes boundary $b{=}L{-}K$ in the text. Layer numbers are illustrative; L is backbone-dependent.

same model, without launching a second full forward pass over a perturbed image or relying on an external reference model. We describe the branch design (Section 3.1), the masked counterfactual construction (Section 3.2), and the contrastive decoding rule (Section 3.3).

3.1 Internal Branch Design

As shown in Figure 2, SIRA centers on a within-model comparison: each multimodal prediction is compared with an internal reference dominated by textual context and language priors. This avoids mismatches from external verifiers or auxiliary models while keeping inference lightweight. Specifically, SIRA maintains two synchronized decoding branches: a **full branch** that performs standard multimodal inference, and a **counterfactual branch** that uses the same prompt, parameters, and history but is blocked from attending to image-token positions from boundary b onward. To reduce cost, layers 0 to $b-1$ are computed once, and branch separation is applied only in the final K layers. The discrepancy between the two branches thus provides an approximate attribution signal, measuring how much a candidate token relies on visual evidence rather than language priors.

Let $\mathbf{x}_{1:S}$ denote the unified prompt tokens after preprocessing, and let $\mathcal{P}_{\text{img}} \subseteq \{1, \ldots, S\}$ denote the positions occupied by image tokens. At decoding step t, the current sequence is $\mathbf{u}_t = [\mathbf{x}_{1:S}; y_{<t}]$. The set $\mathcal{P}_{\text{img}}$ remains fixed during decoding, since generated tokens are ordinary text tokens. We split the L-layer backbone at boundary $b = L - K$, where K denotes the number of post-boundary layers that participate in branch separation. Let $F_{0:b-1}$ denote the shared prefix layers and $F_{b:L-1}$ denote the post-boundary layers. SIRA first obtains a shared boundary representation $\mathbf{h}_t^b = F_{0:b-1}(\mathbf{u}_t)$. The two branches are then defined as two continuations of the same boundary state:

$$\mathbf{h}_t^{\text{full},L} = F_{b:L-1}(\mathbf{h}_t^b; M_t^{\text{causal}}), \qquad \mathbf{h}_t^{\text{cf},L} = F_{b:L-1}(\mathbf{h}_t^b; M_t^{\text{cf}}). \tag{1}$$

The boundary balances two requirements: a shared prefix deep enough to form a stable multimodal interpretation, and enough post-boundary layers to remove visual evidence before it is fully absorbed into text-token representations. Following the staged fusion pattern (Section 2.1), we place b just before the main consolidation stage; Section 5.1 confirms this empirically.

The shared prefix is what makes the contrast interpretable. Rather than comparing the multimodal model to an independently constructed text-only or corrupted-image pass, SIRA compares two continuations of the same internal state: both branches inherit identical prompt processing, decoding history, positional encoding, and early semantic grounding, and the only deliberate intervention after b is whether direct image-token access remains available. The resulting logit discrepancy $\mathbf{z}_t^{\text{full}} - \mathbf{z}_t^{\text{cf}}$ therefore reflects the post-boundary visual contribution rather than unrelated representation mismatch; residual drift from masking is empirically small (Section 5.2). The next subsection formalizes the post-boundary visual-access constraint with masked attention.

3.2 Counterfactual Branch Construction

The counterfactual branch must suppress further visual access while preserving the model's linguistic preference and textual reasoning path. SIRA does not alter the prompt, parameters, or output head; it only removes the post-boundary route through which image tokens can still influence the decoder, namely attention to image-token positions. This mask-level intervention keeps the branch on the

same model dynamics as the full branch and avoids a second full forward pass over a corrupted input, yielding a cleaner counterfactual than input-space perturbations.

Let q and k index positions in the current sequence $\mathbf{u}_t$, and let $M_t^{\text{causal}}(q,k)$ denote standard causal-and-padding validity. The counterfactual mask removes image-token access by excluding image positions both as keys and as query rows in the post-boundary branch, so that image-token states can neither be read by other queries nor produce queries that feed the text stream. Formally, we define counterfactual validity as

$$M_t^{\text{cf}}(q,k) = \begin{cases} 1, & \text{if } M_t^{\text{causal}}(q,k) = 1,\ q \notin \mathcal{P}_{\text{img}},\ k \notin \mathcal{P}_{\text{img}}, \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

where $\mathcal{P}_{\text{img}}$ denotes the image-token positions in the original prompt; generated tokens are text positions. The validity mask is converted into the model's additive attention mask, where valid entries receive 0 and invalid entries receive a large negative value. The mask therefore acts in two phases. In the post-boundary prefix pass over layers $b, \ldots, L-1$ on the original prompt, the $q \notin \mathcal{P}_{\text{img}}$ condition is active and non-trivial: it removes image tokens as query rows, so image-token states are prevented from producing queries that read the prompt and from writing into the post-boundary KV cache. During per-step autoregressive decoding, every new query is a generated text token, so the $q \notin \mathcal{P}_{\text{img}}$ condition is satisfied automatically and only the $k \notin \mathcal{P}_{\text{img}}$ condition remains active, preventing new tokens from attending back to image-token keys. As a result, after boundary b, the counterfactual branch keeps the shared semantic prefix but cannot further use image-token positions, making it a language-prior-dominated internal reference.

Computationally, this design avoids an extra full forward pass by sharing the pre-boundary layers and branching only after a mid-layer split. If a standard decoding step costs L layers, SIRA adds only the post-boundary K layers for the counterfactual stream, giving an idealized per-token cost ratio of $(L+K)/L = 1 + K/L$; for a middle split this is roughly $1.5\times$ rather than the $\sim 2\times$ cost of a second full-model pass. Appendix D reports the corresponding latency measurements.

From layer b onward, the two branches are evaluated in parallel under branch-specific masks. The computation is explicitly batched so that the model compares aligned states at every decoding step without changing the underlying backbone. We preserve positional consistency by duplicating the model's native positional encoding inputs across the two branches. Thus, the only intended difference between the two streams is visual accessibility after the split. With the two branches aligned in this way, decoding can directly compare their token-score distributions.

3.3 Internal Contrastive Decoding

With the two branches aligned, SIRA contrasts their logits at each step. At decoding step t, let $\mathbf{h}_t^{\text{full},L}, \mathbf{h}_t^{\text{cf},L} \in \mathbb{R}^{|\mathbf{u}_t| \times d}$ denote the final-layer hidden states of the full and counterfactual branches for the current sequence $\mathbf{u}_t$, and let $(\cdot)_{\text{last}}$ denote the row at the last sequence position $|\mathbf{u}_t|$, i.e., the representation that drives next-token prediction. The two branch-wise logits are

$$\mathbf{z}_t^{\text{full}} = W_{\text{out}}\,\text{Norm}(\mathbf{h}_t^{\text{full},L})_{\text{last}}, \qquad \mathbf{z}_t^{\text{cf}} = W_{\text{out}}\,\text{Norm}(\mathbf{h}_t^{\text{cf},L})_{\text{last}}, \tag{3}$$

where $\text{Norm}(\cdot)$ and W_{out} denote the model's native final normalization and output projection, shared by both branches. SIRA compares these two predictions in logit space so that tokens supported by the image are amplified relative to those that remain strong under the counterfactual reference. We define the internal contrast $\Delta_t(v) = z_t^{\text{full}}(v) - z_t^{\text{cf}}(v)$ for vocabulary index v and form the decoding logit

$$z_t^{\text{cd}}(v) = z_t^{\text{full}}(v) + \alpha\,\Delta_t(v) = (1+\alpha) z_t^{\text{full}}(v) - \alpha z_t^{\text{cf}}(v), \tag{4}$$

where $\alpha \geq 0$ controls contrast strength. When $\alpha = 0$, the rule reduces to standard decoding; as α increases, tokens that remain strong in the counterfactual branch receive a larger penalty, while tokens whose advantage comes specifically from visual grounding are emphasized. This operation is best understood as approximate baseline subtraction in logit space: $\mathbf{z}_t^{\text{cf}}$ serves as a language-prior-dominated reference constructed in Section 3.2, and subtracting it from $\mathbf{z}_t^{\text{full}}$ yields a proxy for the incremental contribution of visual evidence to each token score. The resulting contrast counteracts language-template bias in real time while preserving the original model parameters and input.

The next token is selected greedily from the contrastive logits, $y_t = \arg\max_{v \in \mathcal{V}} z_t^{\text{cd}}(v)$. The decoding loop maintains a standard full-branch cache alongside a late-layer branch cache for the

stacked two-stream computation: after prefill, each step updates the full branch, then the late-layer parallel branches, and finally applies contrastive fusion to select the next token. Together, the internal branching (Section 3.1), masked counterfactual construction (Section 3.2), and decoding rule above form a training-free, token-level control mechanism that operates within a single model and avoids any second full forward pass.

4 Experiments

4.1 Experimental Setup

Datasets and Metrics. We evaluate SIRA on three complementary hallucination benchmarks that together cover discriminative question answering, short captioning, and open-ended generation.

POPE [21] probes object-level hallucination through balanced Yes/No questions about the presence of specific objects in an image. Each split is 50% positive and 50% negative queries, and three negative-sampling strategies control how strongly textual co-occurrence biases the model toward the wrong answer. Following the standard POPE setup, queries are drawn from MSCOCO [22], A-OKVQA [28], and GQA [12], and we report Accuracy and F1 on each (dataset, split) pair.

CHAIR [27] evaluates object hallucination in free-form captioning by checking whether the objects mentioned in a generated caption actually appear in the image. We report the sentence-level rate CHAIRs and the instance-level rate CHAIRi (lower is better) on a random subset of 500 images from the COCO 2014 validation set, at two generation lengths, Len64 and Len512.

AMBER [36] is a multidimensional captioning benchmark; we use its generation task and report four metrics. CHAIR, Hal, and Cog capture different aspects of hallucinated content (lower is better), while Cover measures descriptive coverage of the image (higher is better). Reporting Hal and Cog jointly with Cover lets us check whether a method reduces hallucination by shortening or hedging the caption rather than by correcting the offending tokens.

Baselines. We adopt LLaVA-v1.5-7B [24] and Qwen2.5-VL-7B [3] as our base LVLMs. We compare SIRA against inference-time methods that mitigate hallucinations by intervening at the decoding stage: DoLa [6], OPERA [11], VCD [18], ICT [5], MaskCD [7], and SID [13]. Appendix C summarizes their venues, intervention mechanisms, and how they differ from SIRA. Implementation details can be seen in Appendix A.

4.2 POPE

Table 1 reports the POPE results. SIRA achieves the best result on all 36 metrics across two backbones, three datasets, and three negative-sampling protocols. Averaged over all POPE settings, SIRA improves Qwen2.5-VL by $+5.4$ accuracy and $+6.7$ F1, and LLaVA-v1.5 by $+4.9$ accuracy and $+6.0$ F1 over their base decoders. It also consistently outperforms the strongest non-SIRA baseline in every setting, with average margins of $+1.23$ points on Qwen2.5-VL and $+0.63$ points on LLaVA-v1.5. The gains extend beyond accuracy: F1 improves across all splits, indicating that SIRA better balances rejecting absent objects and preserving correct positive predictions, rather than simply biasing the model toward negative answers. The advantage is stronger under greater language-prior pressure. On GQA, where visually absent but textually plausible objects make negative questions harder, Qwen2.5-VL improves from 80.9% to 88.4% accuracy on the Popular split and from 78.8% to 84.3% on the Adversarial split. LLaVA-v1.5 shows the same trend, increasing from 78.2% to 82.9% and from 75.1% to 79.6%, respectively. This suggests that SIRA is especially effective when hallucinations arise from object co-occurrence priors rather than random decoding errors. While MaskCD and ICT are competitive in some settings, SIRA consistently surpasses them without external image perturbations, auxiliary intervention vectors, or duplicate full forward passes.

We further examine whether the POPE gains result from a simple Yes/No response-prior shift. Since POPE uses balanced positive and negative questions, uniformly changing the Yes-rate may improve one error type while hurting the other. SIRA does not exhibit this behavior: on LLaVA-v1.5 A-OKVQA-Adversarial and GQA-Adversarial, where the base decoder already has Yes-rates above 50%, SIRA further increases the Yes-rate by $+5.9$ and $+4.3$ points, yet accuracy still improves by $+3.3$ and $+4.5$ points and F1 by $+4.3$ and $+5.1$ points. Thus, the improvements cannot be explained

Dataset	Setting		Qwen2.5-VL								LLaVA-v1.5							
			Base	DoLa	OPERA	VCD	ICT	MaskCD	SID	SIRA	Base	DoLa	OPERA	VCD	ICT	MaskCD	SID	SIRA
COCO	Random	Acc	85.23	86.53	87.31	88.63	87.53	90.05	87.90	**91.13**	83.29	85.97	89.20	87.73	89.18	88.55	88.05	**89.27**
		F1	83.09	84.44	86.92	87.81	85.84	89.60	86.73	**90.49**	81.33	86.14	88.81	87.16	88.48	88.30	87.34	**89.18**
	Popular	Acc	84.53	85.93	87.44	87.12	86.76	88.65	86.90	**89.70**	81.88	82.93	86.64	85.38	86.07	86.25	85.83	**86.83**
		F1	82.58	84.41	86.68	86.40	84.94	88.45	86.00	**89.12**	80.06	83.80	84.89	85.06	86.40	86.00	85.44	**86.51**
	Adversarial	Acc	83.37	83.47	84.78	84.26	86.16	86.05	83.83	**87.83**	78.96	77.17	81.24	80.88	83.30	81.80	81.41	**84.07**
		F1	81.57	81.51	83.45	83.90	84.32	86.15	83.03	**87.39**	77.57	79.41	81.38	81.33	82.54	82.10	81.54	**83.41**
A-OKVQA	Random	Acc	86.40	87.40	88.19	89.22	88.96	90.55	88.49	**92.10**	83.45	83.23	88.03	86.15	89.00	87.45	86.47	**89.13**
		F1	85.07	86.46	88.43	89.01	87.90	90.60	87.93	**91.93**	82.56	84.83	87.00	86.34	88.71	87.55	86.52	**89.52**
	Popular	Acc	85.77	87.53	87.91	87.85	87.43	89.05	87.62	**89.67**	79.90	76.47	83.22	81.85	83.40	83.05	82.48	**83.77**
		F1	84.49	86.06	87.13	87.81	86.39	89.15	87.41	**89.70**	79.59	79.86	84.15	82.82	84.45	83.75	83.20	**84.73**
	Adversarial	Acc	80.37	81.83	80.82	81.27	83.60	82.75	80.84	**83.63**	74.04	68.03	73.82	74.97	75.56	75.90	75.50	**77.30**
		F1	79.68	81.33	81.54	82.38	83.02	83.95	81.51	**84.61**	75.15	74.49	77.91	77.73	78.68	78.40	77.94	**79.43**
GQA	Random	Acc	85.10	87.13	86.02	85.59	88.96	89.25	84.86	**92.23**	83.73	83.70	88.13	86.65	89.23	87.85	86.97	**89.47**
		F1	83.87	86.05	85.29	85.33	87.89	89.10	84.25	**92.09**	82.95	85.29	88.91	86.99	89.34	88.25	87.17	**89.53**
	Popular	Acc	80.87	82.53	81.97	81.83	86.43	86.35	81.61	**88.37**	78.17	74.03	79.27	80.73	80.86	82.05	81.36	**82.93**
		F1	80.33	82.35	82.12	82.23	85.47	86.70	81.83	**88.60**	78.37	78.75	82.11	82.24	82.64	83.10	82.62	**83.75**
	Adversarial	Acc	78.77	82.00	80.24	80.01	84.10	83.25	79.58	**84.27**	75.08	68.73	75.00	76.09	77.40	77.10	76.62	**79.60**
		F1	78.56	81.51	80.64	80.75	83.53	84.30	79.88	**85.18**	76.06	74.78	78.71	78.78	80.11	79.30	78.99	**81.17**

Table 1: POPE results on COCO, A-OKVQA, and GQA. Bold denotes the best result, underline denotes the second-best result, and SIRA column is highlighted.

by a uniform shift toward Yes or No responses, but are instead consistent with selective correction of visually grounded object decisions. Appendix E provides the full per-split decomposition.

4.3 CHAIR

Table 2 reports the CHAIR results on captioning. SIRA achieves the best results across all CHAIR settings, reducing both sentence-level and instance-level hallucination for the two backbones and generation lengths. On Qwen2.5-VL, CHAIRs drops from 20.8 to 16.4 at Len64 and from 35.5 to 31.2 at Len512; on LLaVA-v1.5, it drops from 31.5 to 23.1 and from 50.6 to 43.6, respectively. CHAIRi decreases jointly, e.g., from 17.8 to 14.0 on Qwen2.5-VL and from 20.3 to 15.1 on LLaVA-v1.5 at Len64, indicating that the gains do not come from simply omitting objects but from reducing spurious object mentions. The improvements remain consistent at Len512, suggesting that the internal contrast remains effective when hallucinations accumulate during longer autoregressive generation.

4.4 AMBER

Table 2 reports the AMBER results. SIRA is best on all 8 metrics and, crucially, *reduces hallucination while preserving or raising coverage*: on LLaVA-v1.5 Cover rises from 50.7 to 53.7 and Hal drops from 33.7 to 21.6, while on Qwen2.5-VL Cover is preserved ($47.2 \rightarrow 47.8$) as Hal falls from 23.6 to 20.2. This joint behavior is diagnostic, because other strong baselines obtain their Hal reductions at the expense of coverage: on LLaVA-v1.5, OPERA and VCD both drop Cover from 50.7 to 49.6 while lowering Hal, a pattern consistent with over-hedging or shortening the response rather than correcting the specific tokens that hallucinate.

The Cog metric, which captures cognitive hallucinations that are most tightly coupled to language priors, drops most sharply under SIRA ($3.8 \rightarrow 1.8$ on LLaVA-v1.5 and $1.9 \rightarrow 0.9$ on Qwen2.5-VL, each roughly halving). Because Cog isolates fabrications the model produces even when it is not certain the concept is in the image, its disproportionate reduction is precisely what we would expect from a mechanism that suppresses tokens driven by linguistic co-occurrence rather than by post-boundary visual evidence. SIRA thus removes the error type most aligned with its design target while preserving descriptive content, which we argue is the correct operating point for downstream use where over-hedged captions are as costly as hallucinated ones.

Model	Method	CHAIR				AMBER			
		$C_S^{64}\downarrow$	$C_I^{64}\downarrow$	$C_S^{512}\downarrow$	$C_I^{512}\downarrow$	CHAIR↓	Cover↑	Hal↓	Cog↓
Qwen2.5-VL	Baseline	20.8	17.8	35.5	24.5	5.3	47.2	23.6	1.9
	SID	18.1	15.6	33.0	22.8	4.9	46.3	21.5	1.4
	MaskCD	16.9	14.6	31.8	22.0	4.8	47.4	22.0	1.5
	DoLa	20.1	17.3	35.9	24.6	5.5	46.8	25.1	2.2
	OPERA	18.6	15.9	32.9	22.8	5.1	46.2	21.6	1.7
	VCD	19.3	16.7	33.9	23.3	4.7	45.9	20.5	1.6
	ICT	17.0	14.5	31.9	22.1	4.9	45.1	21.8	1.2
	SIRA (ours)	**16.4**	**14.0**	**31.2**	**21.7**	**4.6**	**47.8**	**20.2**	**0.9**
LLaVA-v1.5	Baseline	31.5	20.3	50.6	30.4	7.3	50.7	33.7	3.8
	SID	26.0	17.2	46.2	27.6	6.1	50.2	27.5	2.8
	MaskCD	24.0	15.7	44.2	26.4	6.6	52.4	31.6	3.0
	DoLa	29.1	21.5	52.4	30.2	7.6	51.6	36.0	4.0
	OPERA	26.7	17.9	46.9	27.8	7.3	49.6	32.0	3.5
	VCD	28.3	19.0	48.7	28.9	6.8	49.6	30.4	3.5
	ICT	24.2	15.9	44.5	26.6	5.2	51.3	23.1	2.1
	SIRA (ours)	**23.1**	**15.1**	**43.6**	**26.0**	**4.8**	**53.7**	**21.6**	**1.8**

Table 2: Main results on CHAIR and AMBER. CHAIR reports C_S/C_I at two caption lengths (64, 512). For AMBER, CHAIR, Hal, and Cog are lower-is-better, while Cover is higher-is-better. Bold denotes the best result and underline denotes the second-best result.

5 Discussion

5.1 Boundary Placement for Internal Reconstruction

SIRA's internal contrast is governed by the split boundary b and contrast strength α. Figure 3 sweeps both on POPE for Qwen2.5-VL and LLaVA-v1.5. Since $b = L - K$, moving b changes the number of post-boundary layers in which the counterfactual branch is blocked from image-token access. Accuracy peaks at a middle split on both backbones and drops when the split is too early or too late. This supports the staged-fusion view in Section 2.1: early splits disrupt multimodal grounding before a stable interpretation forms, whereas late splits leave little post-boundary computation for masking to change the prediction. A mid-layer boundary therefore provides the intended trade-off between preserving a shared multimodal prefix and retaining enough late layers for a meaningful internal contrast. The α sweep shows a similar balance: small α underuses the counterfactual reference, while large α over-amplifies the contrast and may penalize visually compatible tokens. Both backbones peak at $\alpha = 0.5$, indicating that moderate prior subtraction best exposes language-template bias without eroding visually grounded predictions.

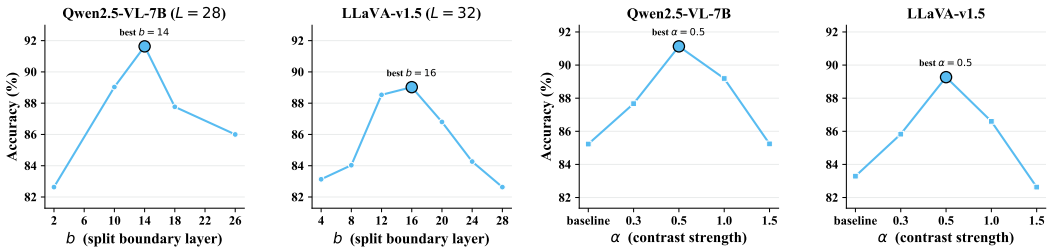


Figure 3: Effect of split boundary b and contrast strength α on POPE accuracy for two models.

5.2 Alignment of the Internal Counterfactual Reference

A useful counterfactual reference should preserve the original semantic trajectory before intervention and diverge primarily where visual access is intentionally removed. Figure 4 shows that SIRA-CF satisfies this criterion better than input-space perturbations. Its layer-wise drift is nearly identical to the full branch before the boundary and grows mainly after the mask is applied, whereas Gaussian blur and patch shuffle drift throughout the network, including early layers that should preserve shared grounding. The output KL is also substantially lower for SIRA-CF (0.012) than for shuffle (0.172) or blur (0.231), while stage-wise statistics still show meaningful late-stage drift. This low-KL/high-late-drift pattern is desirable: a useful reference should not become a globally different generator, since a

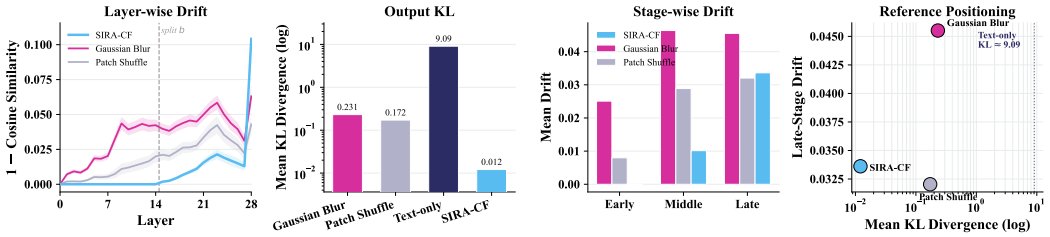


Figure 4: Contrastive reference analysis: (a) layer-wise drift; (b) next-token KL; (c) stage-wise drift; (d) KL versus late-stage drift.

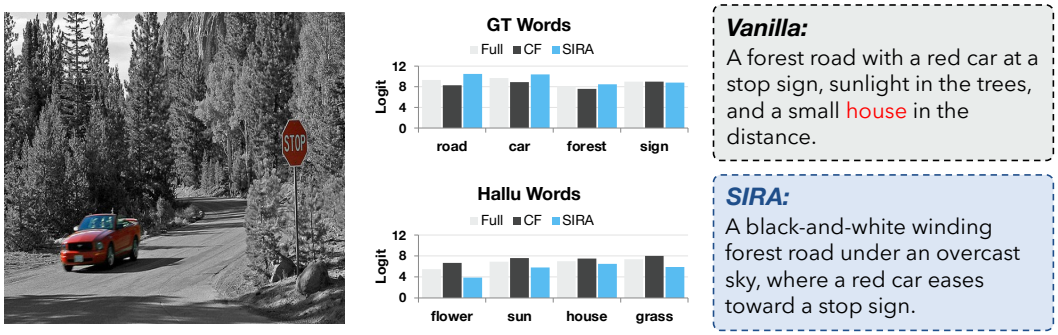


Figure 5: Case study and edge-case analysis on the AMBER benchmark using Qwen2.5-VL. Bars compare Full, CF, and SIRA logits for grounded tokens and hallucinated tokens.

large output KL would make the logit gap reflect distributional mismatch or perturbation artifacts rather than visual attribution. At the same time, it should still deviate in the post-boundary layers where image-token access is removed, so that the resulting discrepancy can expose tokens that rely on continued visual evidence. Thus, SIRA-CF remains comparable to the full branch while separating at the intended intervention site, making it a better positioned internal reference for isolating visual contribution from language-prior bias.

5.3 Contrast Strength and Token-Level Effect

Figure 3 shows the contrast is stable across a moderate range of α, so we next inspect how it changes individual token scores. Figure 5 shows the intended token-level effect of SIRA. For grounded objects, SIRA preserves strong logits under the full visual context. For hallucinated objects in the vanilla response, the counterfactual branch remains competitive, so SIRA lowers the final decoding logit and suppresses the hallucination. The same example also shows a subtle edge case. SIRA removes the unsupported object, but describes the scene as "black-and-white". This phrase is visually motivated because shadows fall across the road and make parts of the surface appear dark or desaturated. However, the road is not intrinsically black-and-white. The remaining issue is an attribute attribution error caused by shadow and lighting, rather than a clear language-prior object hallucination.

6 Conclusion and Limitations

We presented SIRA, a training-free internal contrastive decoder for LVLM hallucination mitigation. By sharing early multimodal computation and branching only in late layers with image-token masking, SIRA constructs an aligned language-prior reference without external perturbations or a second full forward pass. Experiments on POPE, CHAIR, and AMBER show consistent hallucination reduction with preserved coverage. Its main limitations are the remaining late-layer decoding overhead and the requirement for white-box access to hidden states, masks, and caches.

Limitations. Because the post-boundary counterfactual branch is re-evaluated at every decoding step, SIRA also carries a per-step cost above standard decoding that, while well below VCD and OPERA (Table 4), remains non-zero. Finally, constructing the internal contrast requires white-box access to hidden states, attention masks, and cache state, which restricts SIRA to open-weight or otherwise controllable LVLMs rather than black-box APIs.

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A Implementation Details

All methods are evaluated on identical prompts and image inputs, with each baseline run under its original decoding configuration. For SIRA we use $\alpha = 0.5$ with $K = 14$ on Qwen2.5-VL and $K = 16$ on LLaVA-v1.5, applying the counterfactual mask from Section 3.2 without modification; hyperparameters are fixed across datasets. All experiments are conducted on a system equipped with 3 NVIDIA A100 40GB GPUs.

B SIRA Decoding Algorithm

The notation reuses Section 3: $F_{0:b-1}, F_{b:L-1}$ are the shared prefix and post-boundary layer stacks; $M_t^{\text{causal}}, M_t^{\text{cf}}$ are the causal and counterfactual attention masks, where M_t^{cf} blocks every entry whose query row or key column lies in $\mathcal{P}_{\text{img}}$; $\text{Norm}(\cdot)$ and W_{out} are the backbone's final normalization and output projection. We additionally write $C^*_{a:b}$ for branch $*$'s key/value cache over layers $a, \ldots, b-1$, and $F_{a:b}(\cdot;\, C, M)$ for the cached forward that consumes C, appends new key/value rows, and returns the resulting layer-b hidden states together with the updated cache. Lines 1–2 share the prefix cache between branches without recomputation; lines 3–4 build the two prompt-time masks of shape $S \times S$; lines 5–8 perform the late-layer prefill that yields the first generated token y_1. The decoding loop runs the shared layers $F_{0:b-1}$ *once* per step on the previous token (line 11) and forks only at layer b, so SIRA's per-step overhead is K layers of attention and MLP on the last token, giving a latency ratio of approximately $1 + K/L$ rather than the $2\times$ of VCD (Table 4).

Two implementation notes are worth recording. First, the contrasts in lines 7 and 15 are computed in float32 to eliminate half-precision rank reversals on the tail of the vocabulary when the backbone runs in bfloat16. Second, the CF masking *rule*—block keys (and, during prefill, also queries) at $\mathcal{P}_{\text{img}}$ in the CF branch only—is fixed at request time, so only one mask row is appended per decoding step (line 10) and no per-request mask reconstruction is needed. No adaptive plausibility constraint or

Algorithm 1 SIRA: Single-Pass Internal Contrastive Decoding

Require: LVLM with shared prefix layers $F_{0:b-1}$, post-boundary layers $F_{b:L-1}$, normalization $\mathrm{Norm}(\cdot)$, output projection W_{out}; prompt $\mathbf{x}_{1:S}$; image-token positions $\mathcal{P}_{\text{img}} \subseteq \{1, \dots, S\}$; boundary $b = L - K$; strength $\alpha \geq 0$; max length T

Ensure: generated sequence $\mathbf{y}$

$\mathbf{h}_1^b,\ C_{0:b}^{\text{shared}} \leftarrow F_{0:b-1}(\mathbf{x}_{1:S})$ ▷ shared prefix, single pass

$C_{0:b}^{\text{full}},\ C_{0:b}^{\text{cf}} \leftarrow C_{0:b}^{\text{shared}}$

$M_1^{\text{causal}} \leftarrow$ standard causal mask of shape $S \times S$

$M_1^{\text{cf}} \leftarrow M_1^{\text{causal}}$ with rows $q \in \mathcal{P}_{\text{img}}$ and columns $k \in \mathcal{P}_{\text{img}}$ set to $-\infty$

$\mathbf{h}_1^{\text{full},L},\ C_{b:L}^{\text{full}} \leftarrow F_{b:L-1}\big(\mathbf{h}_1^b;\ M_1^{\text{causal}}\big)$

$\mathbf{h}_1^{\text{cf},L},\ C_{b:L}^{\text{cf}} \leftarrow F_{b:L-1}\big(\mathbf{h}_1^b;\ M_1^{\text{cf}}\big)$

$z_1^{\text{full}} \leftarrow W_{\text{out}}\,\mathrm{Norm}\Big(\mathbf{h}_1^{\text{full},L}\Big)_{\text{last}};\quad z_1^{\text{cf}} \leftarrow W_{\text{out}}\,\mathrm{Norm}\Big(\mathbf{h}_1^{\text{cf},L}\Big)_{\text{last}}$ ▷ float32

$z_1^{\text{cd}} \leftarrow (1{+}\alpha)\, z_1^{\text{full}} - \alpha\, z_1^{\text{cf}};\quad y_1 \leftarrow \arg\max_{v \in \mathcal{V}}\ z_1^{\text{cd}}(v);\quad \mathbf{y} \leftarrow (y_1)$

for $t = 2, \dots, T$ **do**

  extend $M_t^{\text{causal}}, M_t^{\text{cf}}$ by one new query row at position $S{+}t{-}1$; in M_t^{cf} set columns $k \in \mathcal{P}_{\text{img}}$ to $-\infty$

  $\mathbf{h}_t^b,\ C_{0:b}^{\text{full}} \leftarrow F_{0:b-1}\big(y_{t-1};\ C_{0:b}^{\text{full}}\big)$ ▷ shared, single-token forward

  $C_{0:b}^{\text{cf}} \leftarrow C_{0:b}^{\text{full}}$ ▷ shared layers stay identical

  $\mathbf{h}_t^{\text{full},L},\ C_{b:L}^{\text{full}} \leftarrow F_{b:L-1}\big(\mathbf{h}_t^b;\ C_{b:L}^{\text{full}}, M_t^{\text{causal}}\big)$

  $\mathbf{h}_t^{\text{cf},L},\ C_{b:L}^{\text{cf}} \leftarrow F_{b:L-1}\big(\mathbf{h}_t^b;\ C_{b:L}^{\text{cf}}, M_t^{\text{cf}}\big)$

  $z_t^{\text{full}} \leftarrow W_{\text{out}}\,\mathrm{Norm}\Big(\mathbf{h}_t^{\text{full},L}\Big)_{\text{last}};\quad z_t^{\text{cf}} \leftarrow W_{\text{out}}\,\mathrm{Norm}\Big(\mathbf{h}_t^{\text{cf},L}\Big)_{\text{last}}$ ▷ float32

  $z_t^{\text{cd}} \leftarrow (1{+}\alpha)\, z_t^{\text{full}} - \alpha\, z_t^{\text{cf}};\quad y_t \leftarrow \arg\max_{v \in \mathcal{V}}\ z_t^{\text{cd}}(v);\quad$ append y_t to $\mathbf{y}$

  if y_t is the end-of-sequence token **then break**

  end if

end for

return $\mathbf{y}$

vocabulary truncation is applied: as argued in Section 2.2, the CF distribution already lives on the base model's token manifold.

C Baseline Details and Comparison

We compare against representative inference-time hallucination mitigation methods. DoLa [6] contrasts logits from different transformer layers to favor predictions that are more strongly supported by mature representations. VCD [18] constructs a visual contrast by comparing the original image with a distorted visual input, such as a perturbed or degraded image. OPERA [11] penalizes over-trusted attention patterns and uses retrospection allocation during generation. ICT [5] introduces image-object cross-level trusted intervention to steer decoding with object-level signals. MaskCD [7] masks image heads to form a contrastive decoding signal. SID [13] uses self-introspective decoding to construct an additional reference for hallucination reduction.

Table 3 summarizes method-level properties and trade-offs rather than numerical performance. We organize the comparison around four practical questions: whether the method is tailored to LVLMs, preserves the original visual input, avoids external tools/models, and keeps decoding overhead low. Several baselines meet part of these requirements, while SIRA satisfies all four.

D Efficiency Analysis

SIRA's efficiency follows directly from shared-prefix branching. A standard autoregressive step evaluates all L layers once. SIRA evaluates the same full branch while adding only a counterfactual continuation through the final K post-boundary layers, so the per-step compute scales as $L + K$ layers and the idealized ratio is $(L + K)/L = 1 + K/L$, rather than two full L-layer passes. Our measurements below focus on wall-clock latency.

Method	LVLM-specific	Preserves input	No external tools/models	Low decoding overhead
DoLa [6]	✗	✓	✓	✓
VCD [18]	✓	✗	✓	✗
OPERA [11]	✓	✓	✓	✗
ICT [5]	✓	✓	✗	✓
MaskCD [7]	✓	✓	✓	✗
SID [13]	✓	✓	✓	✗
SIRA (ours)	✓	✓	✓	✓

Table 3: Property comparison of representative inference-time hallucination mitigation methods. Low decoding overhead means avoiding extra full reference passes, self-introspective reference construction, or beam-search retrospection.

Table 4 quantifies the practical overhead. SIRA adds 1.4–$1.5\times$ overhead over standard decoding, compared with $\sim 2.1\times$ for VCD, which requires a full duplicate pass on a perturbed image, and 5–$6\times$ for OPERA's beam-search manipulation. The ratio remains stable as generation length grows from 32 to 128 tokens, because the per-step cost is dominated by late-layer branching rather than by a second full multimodal trajectory.

Method	32-Token Len		64-Token Len		128-Token Len	
	ms	Rel.	ms	Rel.	ms	Rel.
Qwen2.5-VL	587	$\times 1.0$	1000	$\times 1.0$	1868	$\times 1.0$
VCD	1205	$\uparrow \times 2.1$	2063	$\uparrow \times 2.1$	3847	$\uparrow \times 2.1$
OPERA	3100	$\uparrow \times 5.3$	7786	$\uparrow \times 5.8$	10966	$\uparrow \times 5.9$
SIRA (ours)	**838**	$\uparrow \times 1.4$	**1496**	$\uparrow \times 1.5$	**2781**	$\uparrow \times 1.5$

Table 4: Comparison of inference efficiency across methods on Qwen2.5-VL with varying generation lengths. Inference times are recorded in milliseconds.

E POPE Yes-Rate Decomposition

Table 5 and Table 6 report the full POPE decomposition (Accuracy, F1, Precision, Recall, FPR, Yes-rate) for every (backbone, dataset, split, method) cell. Across both backbones and all nine splits, SIRA's Yes-rate rises relative to the backbone (by $+2.2$ to $+9.8$ points), so Yes-rate shift alone is not the distinguishing quantity. The sharper diagnostic is how Yes-rate moves *with respect to accuracy*: on POPE splits whose baseline Yes-rate already exceeds 50%, a pure Yes/No rebalancing would push the decoder past the balanced optimum and *degrade* accuracy. SIRA exhibits the opposite behavior on every such split: on LLaVA-v1.5 A-OKVQA-Adversarial (baseline Yes 54.5%, SIRA Yes 60.4%) accuracy rises from 74.0 to 77.3, and on LLaVA-v1.5 GQA-Adversarial (baseline Yes 54.1%, SIRA Yes 58.3%) accuracy rises from 75.1 to 79.6. The Yes-rate increase therefore does not capture a unidirectional rebalancing of Yes/No mass but accompanies genuine visually grounded correction.

Dataset	Setting	Method	Acc	F1	Prec. %	Recall %	FPR %	Yes %
COCO	Random	Baseline	85.23	83.09	97.17	72.57	2.11	37.34
		DoLa	86.53	84.44	99.95	73.10	0.04	36.57
		OPERA	87.31	86.92	89.68	84.33	9.71	47.02
		VCD	88.63	87.81	94.64	81.90	4.64	43.27
		ICT	87.53	85.84	99.30	75.59	0.53	38.06
		MaskCD	90.05	89.60	93.84	85.72	5.62	45.67
		SID	87.90	86.73	96.01	79.08	3.28	41.18
		SIRA	91.13	90.49	97.53	84.40	2.14	43.27
	Popular	Baseline	84.53	82.58	94.49	73.34	4.28	38.81
		DoLa	85.93	84.41	94.63	76.18	4.32	40.25
		OPERA	87.44	86.68	92.26	81.73	6.85	44.29
		VCD	87.12	86.40	91.52	81.83	7.59	44.71
		ICT	86.76	84.94	98.48	74.68	1.16	37.92
		MaskCD	88.65	88.45	90.04	86.92	9.62	48.27
		SID	86.90	86.00	92.34	80.47	6.67	43.57
		SIRA	89.70	89.12	94.44	84.37	4.97	44.67
	Adversarial	Baseline	83.37	81.57	91.47	73.60	6.86	40.23
		DoLa	83.47	81.51	92.47	72.87	5.93	39.40
		OPERA	84.78	83.45	91.44	76.74	7.18	41.96
		VCD	84.26	83.90	85.86	82.02	13.50	47.76
		ICT	86.16	84.32	97.25	74.43	2.11	38.27
		MaskCD	86.05	86.15	85.54	86.77	14.67	50.72
		SID	83.83	83.03	87.35	79.12	11.46	45.29
		SIRA	87.83	87.39	90.67	84.34	8.68	46.51
A-OKVQA	Random	Baseline	86.40	85.07	94.29	77.49	4.69	41.09
		DoLa	87.40	86.46	93.43	80.46	5.66	43.06
		OPERA	88.19	88.43	86.67	90.26	13.88	52.07
		VCD	89.22	89.01	90.78	87.31	8.87	48.09
		ICT	88.96	87.90	97.24	80.20	2.28	41.24
		MaskCD	90.55	90.60	90.12	91.08	9.98	50.53
		SID	88.49	87.93	92.43	83.85	6.87	45.36
		SIRA	92.10	91.93	93.95	89.99	5.79	47.89
	Popular	Baseline	85.77	84.49	92.84	77.52	5.98	41.75
		DoLa	87.53	86.06	97.56	76.98	1.92	39.45
		OPERA	87.91	87.13	93.14	81.85	6.03	43.94
		VCD	87.85	87.81	88.10	87.52	11.82	49.67
		ICT	87.43	86.39	94.18	79.79	4.93	42.36
		MaskCD	89.05	89.15	88.34	89.97	11.87	50.92
		SID	87.62	87.41	88.92	85.95	10.71	48.33
		SIRA	89.67	89.70	89.44	89.96	10.62	50.29
	Adversarial	Baseline	80.37	79.68	82.58	76.97	16.23	46.60
		DoLa	81.83	81.33	83.63	79.15	15.49	47.32
		OPERA	80.82	81.54	78.59	84.72	23.08	53.90
		VCD	81.27	82.38	77.77	87.57	25.03	56.30
		ICT	83.60	83.02	86.06	80.18	12.98	46.58
		MaskCD	82.75	83.95	78.49	90.23	24.73	57.48
		SID	80.84	81.51	78.76	84.46	22.78	53.62
		SIRA	83.63	84.61	79.83	90.00	22.74	56.37
GQA	Random	Baseline	85.10	83.87	91.42	77.47	7.27	42.37
		DoLa	87.13	86.05	93.93	79.39	5.13	42.26
		OPERA	86.02	85.29	89.99	81.06	9.02	45.04
		VCD	85.59	85.33	86.90	83.82	12.64	48.23
		ICT	88.96	87.89	97.32	80.12	2.20	41.16
		MaskCD	89.25	89.10	90.36	87.87	9.37	48.62
		SID	84.86	84.25	87.79	80.99	11.27	46.13
		SIRA	92.23	92.09	93.78	90.46	6.00	48.23
	Popular	Baseline	80.87	80.33	82.66	78.12	16.38	47.25
		DoLa	82.53	82.35	83.21	81.51	16.45	48.98
		OPERA	81.97	82.12	81.44	82.81	18.87	50.84
		VCD	81.83	82.23	80.46	84.08	20.42	52.25
		ICT	86.43	85.47	91.98	79.82	6.96	43.39
		MaskCD	86.35	86.70	84.53	88.98	16.28	52.63
		SID	81.61	81.83	80.86	82.82	19.60	51.21
		SIRA	88.37	88.60	86.88	90.39	13.65	52.02
	Adversarial	Baseline	78.77	78.56	79.34	77.79	20.25	49.02
		DoLa	82.00	81.51	83.79	79.35	15.35	47.35
		OPERA	80.24	80.64	79.04	82.31	21.83	52.07
		VCD	80.01	80.75	77.87	83.85	23.83	53.84
		ICT	84.10	83.53	86.64	80.64	12.44	46.54
		MaskCD	83.25	84.30	79.33	89.94	23.44	56.69
		SID	79.58	79.88	78.72	81.07	21.91	51.49
		SIRA	84.27	85.18	80.52	90.41	21.87	56.14

Table 5: POPE decomposition on Qwen2.5-VL. SIRA's Yes-rate rises on every split, but accuracy rises jointly even on Yes-biased splits where a pure Yes/No rebalancing would predict accuracy loss.

Dataset	Setting	Method	Acc	F1	Prec. %	Recall %	FPR %	Yes %
COCO	Random	Baseline	83.29	81.33	92.14	72.79	6.21	39.50
		DoLa	85.97	86.14	85.11	87.20	15.26	51.23
		OPERA	89.20	88.81	92.14	85.71	7.31	46.51
		VCD	87.73	87.16	91.41	83.29	7.83	45.56
		ICT	89.18	88.48	94.60	83.10	4.74	43.92
		MaskCD	88.55	88.30	90.27	86.41	9.31	47.86
		SID	88.05	87.34	92.86	82.44	6.34	44.39
		SIRA	89.27	89.18	89.93	88.44	9.90	49.17
	Popular	Baseline	81.88	80.06	89.00	72.75	8.99	40.87
		DoLa	82.93	83.80	79.74	88.30	22.44	55.37
		OPERA	86.64	84.89	97.69	75.06	1.78	38.42
		VCD	85.38	85.06	86.96	83.24	12.48	47.86
		ICT	86.07	86.40	84.40	88.50	16.36	52.43
		MaskCD	86.25	86.00	87.59	84.46	11.96	48.21
		SID	85.83	85.44	87.86	83.15	11.49	47.32
		SIRA	86.83	86.51	88.66	84.46	10.80	47.63
	Adversarial	Baseline	78.96	77.57	83.06	72.76	14.84	43.80
		DoLa	77.17	79.41	72.31	88.05	33.71	60.88
		OPERA	81.24	81.38	80.78	81.99	19.51	50.75
		VCD	80.88	81.33	79.46	83.29	21.53	52.41
		ICT	83.30	82.54	86.48	78.95	12.35	45.65
		MaskCD	81.80	82.10	80.77	83.48	19.88	51.68
		SID	81.41	81.54	80.97	82.11	19.29	50.70
		SIRA	84.07	83.41	87.02	80.09	11.95	46.02
A-OKVQA	Random	Baseline	83.45	82.56	87.25	78.35	11.45	44.90
		DoLa	83.23	84.83	77.44	93.78	27.32	60.55
		OPERA	88.03	87.00	95.19	80.11	4.05	42.08
		VCD	86.15	86.34	85.17	87.54	15.24	51.39
		ICT	89.00	88.71	91.11	86.43	8.43	47.43
		MaskCD	87.45	87.55	86.86	88.25	13.35	50.80
		SID	86.47	86.52	86.20	86.84	13.90	50.37
		SIRA	89.13	89.52	86.42	92.85	14.59	53.72
	Popular	Baseline	79.90	79.59	80.84	78.38	18.58	48.48
		DoLa	76.47	79.86	69.80	93.30	40.36	66.83
		OPERA	83.22	84.15	79.73	89.09	22.65	55.87
		VCD	81.85	82.82	78.62	87.50	23.80	55.65
		ICT	83.40	84.45	79.43	90.15	23.35	56.75
		MaskCD	83.05	83.75	80.43	87.36	21.26	54.31
		SID	82.48	83.20	79.92	86.77	21.81	54.29
		SIRA	83.77	84.73	80.00	90.06	22.52	56.29
	Adversarial	Baseline	74.04	75.15	72.07	78.51	30.43	54.47
		DoLa	68.03	74.49	61.97	93.35	57.29	75.32
		OPERA	73.82	77.91	67.38	92.34	44.70	68.52
		VCD	74.97	77.73	70.01	87.36	37.42	62.39
		ICT	75.56	78.68	69.77	90.19	39.07	64.63
		MaskCD	75.90	78.40	71.03	87.47	35.67	61.57
		SID	75.50	77.94	70.88	86.56	35.56	61.06
		SIRA	77.30	79.43	72.62	87.65	33.05	60.35
GQA	Random	Baseline	83.73	82.95	87.13	79.16	11.70	45.43
		DoLa	83.70	85.29	77.71	94.51	27.11	60.81
		OPERA	88.13	88.91	83.43	95.16	18.90	57.03
		VCD	86.65	86.99	84.83	89.26	15.96	52.61
		ICT	89.23	89.34	88.44	90.26	11.80	51.03
		MaskCD	87.85	88.25	85.44	91.25	15.55	53.40
		SID	86.97	87.17	85.85	88.53	14.59	51.56
		SIRA	89.47	89.53	89.02	90.04	11.10	50.57
	Popular	Baseline	78.17	78.37	77.66	79.09	22.75	50.92
		DoLa	74.03	78.75	66.64	96.24	48.18	72.21
		OPERA	79.27	82.11	72.22	95.14	36.60	65.87
		VCD	80.73	82.24	76.26	89.23	27.77	58.50
		ICT	80.86	82.64	75.61	91.11	29.39	60.25
		MaskCD	82.05	83.10	78.51	88.26	24.16	56.21
		SID	81.36	82.62	77.39	88.61	25.89	57.25
		SIRA	82.93	83.75	79.91	87.98	22.12	55.05
	Adversarial	Baseline	75.08	76.06	73.18	79.17	29.01	54.09
		DoLa	68.73	74.78	62.66	92.72	55.26	73.99
		OPERA	75.00	78.71	68.54	92.43	42.43	67.43
		VCD	76.09	78.78	70.81	88.77	36.59	62.68
		ICT	77.40	80.11	71.53	91.02	36.22	63.62
		MaskCD	77.10	79.30	72.35	87.73	33.53	60.63
		SID	76.62	78.99	71.72	87.90	34.66	61.28
		SIRA	79.60	81.17	75.37	87.94	28.74	58.34

Table 6: POPE decomposition on LLaVA-v1.5. SIRA's Yes-rate rises on every split, but accuracy rises jointly even on Yes-biased splits where a pure Yes/No rebalancing would predict accuracy loss.

F Additional Case Study

We provide additional logit-level case studies on the AMBER benchmark in Figure 6. Eight cases generated by Qwen2.5-VL are shown under the same caption-mode setting. For each case we display the input image and two logit-delta bar charts: one for ground-truth object tokens (GT Words) and one for the hallucinated tokens emitted by the unmodified backbone (Hallu Words); within each chart the Full and CF bars report the base model's logit under the full multimodal context and under the post-boundary image-masked counterfactual, and the SIRA bar reports the contrastive logit $(1+\alpha)\, z^{\text{full}} - \alpha\, z^{\text{cf}}$ actually used for decoding.

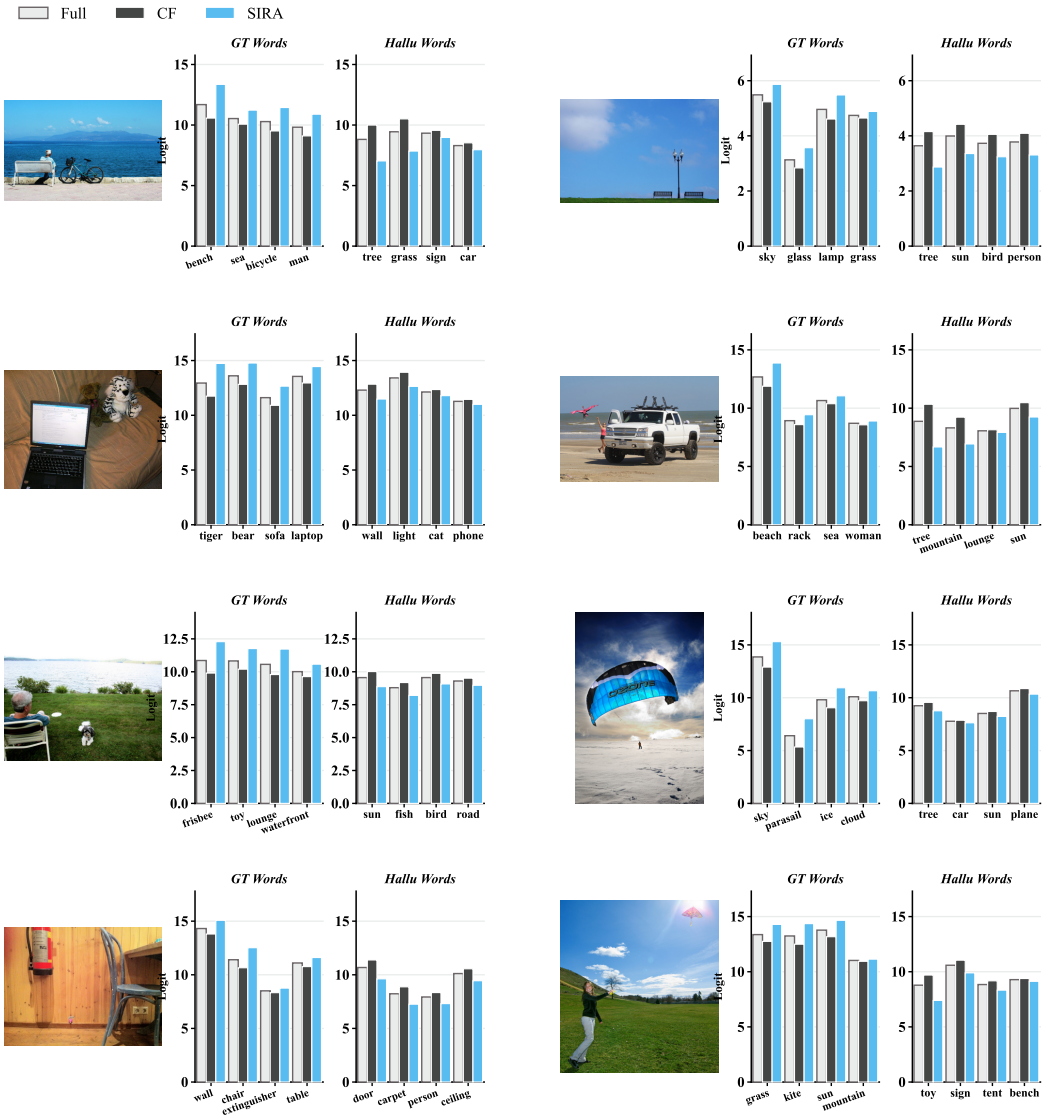


Figure 6: Additional case studies on the AMBER benchmark. Eight AMBER cases are shown; for each case we display the input image, the GT-word bar chart, and the Hallu-word bar chart, comparing the Full, CF, and SIRA logits at the token level.

Three patterns recur across all eight cases and are worth calling out explicitly. (i) On GT Words, the Full bar consistently exceeds CF by a clear margin and SIRA pushes the value further up, so the contrast *reinforces* visually grounded predictions rather than flipping them. (ii) On Hallu Words, CF is comparable to or strictly higher than Full on a non-trivial fraction of tokens—the cleanest

token-level signature of language-prior-driven hallucination, since these tokens remain probable even after the image is masked away. SIRA collapses such tokens to the lowest of the three bars. (iii) On GT Words where Full $\approx$ CF, the SIRA bar barely moves; the contrastive correction is therefore self-gating, leaving already image-independent-but-correct tokens untouched and concentrating its effect on the tokens that actually need it.

These token-level observations are the microscopic counterpart of the POPE Yes-rate decomposition in Appendix E: the *simultaneous* rise of Yes-rate and accuracy on Yes-biased POPE splits is not a uniform Yes/No rebalancing but the aggregate effect of pattern (ii) suppressing prior-driven false positives while pattern (i) preserves true positives. Figure 6 thus links the macroscopic behavior quantified in Section 4.2 and Table 5 to a concrete token-level mechanism.